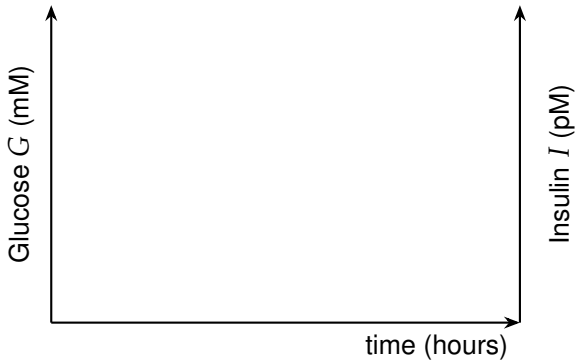
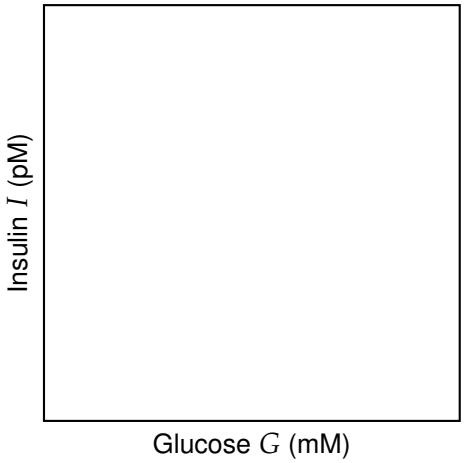
Lecture 6 Live Problems

Insulin–glucose, finished. Parameters. Separation of time scale. Modeling reflection.

Points toward lab manual pages: **IG5, IG6, IG7, IG8** (section L1 S5)

1. Qualitative expectations. Sketch time series for glucose G (mM) and insulin I (pM) in the hours after a large meal. Then sketch the trajectory in glucose–insulin state space.





2. Quantitative expectations. With all parameters m, s, q, b, γ set equal to 1, the change equations for glucose and insulin become

$$\frac{\Delta G}{\Delta t} = 1 - I\,G, \qquad \frac{\Delta I}{\Delta t} = \frac{G^2}{1+G^2} - I.$$

Start with $G = 2$ mM and $I = 1$ pM at time $t = 0$. Predict the future with a table, using a time interval $\Delta t = 1$ hour.

t (hours)	G (mM)	I (pM)	ΔG (mM)	ΔI (pM)
0	2	1		
1				
2				
3				

3. Insulin in the absence of glucose. When $G = 0$, the insulin change equation becomes

$$\frac{\Delta I}{\Delta t} = -\gamma I.$$

Set $\gamma = 0.1$ per hour and start with $I = 3$ pM at time $t = 0$. Predict the future with a table, using a time interval $\Delta t = 1$ hour, and sketch the time series.

t (hours)	I (pM)	ΔI (pM)
0	3	
1		
2		
3		

